

Final 2025: Visual Programming Question Guide

Question 8

How do you create a jagged array that contains strings?

Choices:

- `string[][] jaggedStrings = new string[];`
- `string[] jaggedStrings = new string`
- `[][]string jaggedStrings`
- `string jaggedStrings`

Correct Answer: `string[][] jaggedStrings = new string[];`

Explanation: A jagged array is a one-dimensional array where each element is also a one-dimensional array. Because it is an array of arrays, the syntax requires two sets of brackets. When initializing it, the first set of brackets must specify the number of rows.

Question 9

The ToolTip will always remain visible until the user clicks on the control.

Choices:

- True
- False

Correct Answer: False

Explanation: The visibility duration of a ToolTip is controlled by the `AutoPopDelay` property. This property determines how many milliseconds a ToolTip remains on the screen once it has been displayed, with a default setting of 5,000 milliseconds (5 seconds).

Question 10

You call the SetLastError method with an empty string as the error message.

Choices:

- The error icon is hidden for the specified control
- The error icon is displayed with no message
- An exception is thrown

Correct Answer: The error icon is hidden for the specified control

Explanation: The ErrorProvider component uses the SetLastError method to display a blinking error icon next to a control. The error icon remains visible until SetLastError is called again for that same control using an empty string ("") as the message argument.

Question 11

Which property of a ScrollBar determines the current position of the scroll box?

Choices:

- Position
- Value
- Location
- CurrentScroll

Correct Answer: Value

Explanation: The Value property is an integer that represents the current position of the slider. This value is adjusted as the user moves the slider along the track. Moving a horizontal slider to the right or a vertical slider to the bottom increases this value.

Question 12

Given the following method: public void DisplayMessage(string title, string content, int duration = 5). Which call is invalid?

Choices:

- DisplayMessage("Warning", content: "Low Battery");
- DisplayMessage(title: "Warning", content: "Low Battery", duration: 10);
- DisplayMessage(content: "Low Battery", "Warning", 10);

- `DisplayMessage("Warning", "Low Battery", duration: 10);`

Correct Answer: `DisplayMessage(content: "Low Battery", "Warning", 10);`

Explanation: While C# allows named arguments (e.g., `parameterName: value`), they are generally used to specify which parameter receives an argument regardless of order. However, according to standard C# rules (not explicitly detailed in the sources but implied by conventional usage in), once a named argument is used, all subsequent arguments must also be named. Calling `content:` and then following it with positional arguments like `"Warning"` is invalid.

Question 13

What will the following code output? `string str1 = " Hello "; char str2 = str1.Remove(1); MessageBox.Show(str2.ToString());`

Choices:

- Hello
- H
- Space
- Error

Correct Answer: Error

Explanation: The `Remove` method returns a modified copy of the calling string object, not a single character. Because `char` variables can only hold a single character and are not compatible with strings, you cannot assign the result of `str1.Remove(1)` to `str2`.

Question 14

Which method is used to programmatically add an item to a `ListBox`?

Choices:

- `ListBox.AddItem()`
- `ListBox.Items.Add()`
- `ListBox.Items.Insert()`
- `ListBox.Items.Item.adds`

Correct Answer: `ListBox.Items.Add()`

Explanation: To add an item to a ListBox control at run time through code, you must call the control's Items.Add method.

Question 15

The default value of the AutoSize property of a Label control is False.

Choices:

- True
- False

Correct Answer: False

Explanation: By default, a Label control's AutoSize property is set to True. This causes the label to automatically resize itself to accommodate the size of the text currently being displayed.

Question 16

What will be the output of the following code when the button is clicked twice?

Choices:

- "Button clicked 1 times" on the first click, and "Button clicked 2 times" on the second click.
- A message box showing "Button clicked 1 times" every time.
- Error.

Correct Answer: "Button clicked 1 times" on the first click, and "Button clicked 2 times" on the second click.

Explanation: This behavior occurs if the counter variable is declared as a field. Unlike local variables, which are destroyed when a method ends, a field is declared inside a class but outside any method, allowing all methods to access it and its value to persist for the entire life of the form.

Question 17

Which property of the Label control is used to align the text within the Label?

Choices:

- TextAlign
- Alignment
- TextPosition
- Layout

Correct Answer: TextAlign

Explanation: The TextAlign property determines the position of the text within the label's bounding box. It can be set to various alignments, such as TopLeft, MiddleCenter, or BottomRight.

Question 18

The Load event takes place after the form is displayed on the screen.

Choices:

- True
- False

Correct Answer: False

Explanation: The Load event takes place just before the form is displayed. This allows developers to write event handlers that perform initialization tasks immediately before the user sees the form.

Question 19

Which property of a TabControl contains the collection of tabs?

Choices:

- Tabs
- TabPages
- TabCollection
- Pages

Correct Answer: TabPages

Explanation: A TabControl is a container for TabPage controls. To manage these pages or add new ones, you must access the TabPages collection property.

Question 20

You can use an ImageList control to store images of different sizes.

Choices:

- True
- False

Correct Answer: False

Explanation: All images stored in an ImageList control should be the same size. If the images are not the same size as specified in the ImageSize property, they will appear distorted when displayed.

Question 21

For a button with the Text property set to "&Open", which key combination will activate it?

Choices:

- Alt + O
- Ctrl + O
- Shift + O
- Ctrl + Alt + O

Correct Answer: Alt + O

Explanation: The ampersand (&) identifies the following letter as an access key. Access keys are triggered by pressing the Alt key in combination with the underlined letter.

Question 22

When the button is clicked, what will be the output? (points.X.ToString())

Choices:

- 0
- 5
- 10
- Error

Correct Answer: 0

Explanation: When you use the new operator to create an array of structure instances, the operator not only creates the instances in memory but also initializes the fields with a default value of 0. Since the code only assigned values to points, points.X remains the default value of 0.

Question 23

How can you disable a ToolStripMenuItem in a MenuStrip control?

Choices:

- Set the Enabled property of the ToolStripMenuItem to false
- Set the Visible property of the ToolStripMenuItem to false
- Set the Enabled property of the MenuStrip to false
- Set the Text property of the ToolStripMenuItem to disabled

Correct Answer: Set the Enabled property of the ToolStripMenuItem to false

Explanation: Setting the Enabled property of a menu item to False disables it, causing it to appear dimmed (grayed out) and preventing the user from selecting it.

Question 24

The Text property of a ComboBox always corresponds to the value of the selected item.

Choices:

- True
- False

Correct Answer: False

Explanation: In Drop-Down and Simple combo box styles, users can type their own text into the input area. The Text property contains either the typed input or the selected item, meaning it is not strictly limited to items in the list.

Question 25

Which method is used to generate a random floating-point number between 0.0 and 1.0 (exclusive)?

Choices:

- `random.NextFloat()`
- `random.NextDouble()`
- `random.Double()`
- `random.Float()`

Correct Answer: `random.NextDouble()`

Explanation: The `NextDouble` method of the `Random` class returns a random floating-point number between 0.0 and 1.0, but specifically excludes the value 1.0.

Question 26

You can specify a path as well as a filename in the argument that you pass to the `File.CreateText` method, but not to `File.AppendText` method.

Choices:

- True
- False

Correct Answer: False

Explanation: You can specify both a path and a filename in the argument passed to both the `File.CreateText` and `File.AppendText` methods.

Question 28

In a Windows Forms application, what is the visual indicator of an access key on a button?

Choices:

- Letter is bold
- Underlined
- Highlighted
- Grayed out

Correct Answer: The letter is underlined

Explanation: When a letter is assigned as an access key using an ampersand, it appears underlined on the control to indicate its function.

Question 29

`string str1 = "hello"; string str2 = "HELLO"; String.Compare(str1, str2, true);` The method returns:

Choices:

- 0
- True
- False
- Error

Correct Answer: 0

Explanation: The `String.Compare` method returns 0 if the two strings are equal. Providing `true` as the third argument tells the method to perform a case-insensitive comparison.

Question 30

How can you retrieve an image from the `ImageList`?

Choices:

- `ImageList.GetImage(index)`
- `ImageList.Images[index]`
- `ImageList[index]`
- `ImageList.Image(index)`

Correct Answer: `ImageList.Images[index]`

Explanation: Images in an ImageList are stored in the Images collection property and can be retrieved using their integer index.

Question 31

What will the following code output? `string text = "C# Programming"; string result = text.Substring(3);`

Choices:

- #C
- Programming
- Programming #
- P

Correct Answer: Programming

Explanation: The Substring(start) method returns a string containing characters from the specified starting index to the end of the string. In "C# Programming", index 3 is the letter 'P', so the result is "Programming".

Question 32

You can assign either decimal or int values to decimal variables, but you cannot assign double values to decimal variables.

Choices:

- True
- False

Correct Answer: True

Explanation: While decimal variables are compatible with int and decimal literals, they are not compatible with double values due to differences in precision.

Question 33

Which namespace must be included to use the File.CreateText method?

Choices:

- System.Windows.Forms
- System.IO
- System.Text
- System.Collections

Correct Answer: System.IO

Explanation: Classes involved in file input and output, such as File, StreamWriter, and StreamReader, are located in the System.IO namespace.

Question 34

What will the following code output? `string text = "Hello, World!"; int position = text.IndexOf('o', 5);`

Choices:

- 4
- 8
- 5
- -1

Correct Answer: 8

Explanation: `IndexOf(ch, start)` searches for the first occurrence of a character starting at a specific position. The first 'o' is at index 4, but since the search starts at index 5, it finds the next 'o' at index 8.

Question 35

Which event is typically used to detect when a `RadioButton` has been selected?

Choices:

- Click
- Selected
- CheckChanged
- Checked

Correct Answer: CheckedChanged

Explanation: The CheckedChanged event occurs whenever a RadioButton (or CheckBox) control's Checked property changes.

Question 36

To ensure a TextBox control receives focus first when the form loads, what would you do?

Choices:

- Set its TabIndex property to -1
- Set its TabIndex property to 0
- Set its TabOrder property to -1
- Set its TabOrder property to 0

Correct Answer: Set its TabIndex property to 0

Explanation: The control with a TabIndex of 0 is the first in the tab order and receives focus automatically when the form runs.

Question 37

How can you define multiple delimiters for tokenizing a string using the Split method?

Choices:

- By passing a string containing all delimiters
- By passing a char array of delimiters
- By passing a List<char> of delimiters
- By passing a delimiter string separated by commas

Correct Answer: By passing a char array of delimiters

Explanation: You can call the Split method and pass a char array as an argument, where each element in the array is treated as a delimiter.

Question 38

The AppendText method creates a new file if the specified file does not exist.

Choices:

- True
- False

Correct Answer: True

Explanation: The File.AppendText method preserves existing files to add data, but it acts like File.CreateText by creating the file if it does not already exist.

Question 39

The default behavior of a TextBox is to select all text when it gains focus.

Choices:

- True
- False

Correct Answer: False

Explanation: Standard text boxes do not automatically select all text; this is often a custom behavior implemented by developers using the SelectionStart and SelectionLength properties for user convenience.

Question 40

How can you open a second form as a modal form from the main form?

Choices:

- Show()
- ShowDialog()
- Open()
- Run()

Correct Answer: Form2.ShowDialog()

Explanation: The ShowDialog method displays a form in a modal fashion, meaning the user cannot interact with any other form in the application until the modal form is closed.

Question 41

The property that determines how a background image is displayed on a form is:

Choices:

- BackgroundImageAlignment
- BackgroundImageDisplayMode
- BackgroundImageLayout
- BackgroundImageStyle

Correct Answer: BackgroundImageLayout

Explanation: The BackgroundImageLayout property specifies the layout (e.g., Tile, Center, Stretch, Zoom) used for the background image.

Question 42

Which method do you use to ensure a control retains focus after an event such as a button click?

Choices:

- Control.Focus();
- Control.Select();
- Control.SetFocus();
- Control.GainFocus();

Correct Answer: Control.Focus();

Explanation: The Focus method is the standard way to programmatically move the keyboard focus to a specific control.

Question 43

**What will the following code output? `string Name = "hmed"; Name = 'A';
MessageBox.Show(Name);`**

Choices:

- Ahmed
- hmed
- A
- Error

Correct Answer: Error

Explanation: In C#, subscripts provide read-only access to the characters in a string. You cannot use a subscript to modify a character within a string object; attempting to do so will result in a compiler error.

Question 44

A ComboBox can display multiple columns of data simultaneously.

Choices:

- True
- False

Correct Answer: False

Explanation: A ComboBox displays a single list of items in a single column. Although it is a combination of a text box and a list box, the standard control does not support multiple simultaneous data columns.

Wish you all the best!

Fathy.